

P8X Memory Map

The 64 KB address space after the **6 KB-ROM reclaim**: the firmware ROM gave up its unused top 2 KB, that became a low RAM island for relocated scratch, and the transient program area floor dropped to **\$5900** — **+2 KB** for every program, enough to let the C compiler self-host.

ROM
6 KB \$0000-\$17FF

TPA BASE
\$5900

TPA SIZE
39.8 KB

I/O PAGE
\$FF00

ADDRESS	REGION & CONTENTS	SIZE
\$0000	Firmware ROM ROM Monitor + BIOS. Reset vector \$0000, BIOS jump table \$0100. ~5.2 KB used; shrunk 8 KB→6 KB to open the island below.	6 KB -\$17FF
\$1800	RAM island RAM · reclaimed The 2 KB freed by the smaller ROM, now holding the scratch that used to pin the TPA floor. \$1800 IBUF stdin read-stream buffer 512 B \$1A00 PATHBUF + shell vars PATH, drive & run state 256 B \$1B00 APBUF append / sh sector · tab-complete (overlay) 512 B \$1D00 SBUF monitor sector buffer 512 B \$1F00 BIOS/FS scratch FNAME · LBA · DIRLBA · flags 256 B	2 KB -\$1FFF
\$2000	OS image RAM Shell + filesystem + the resident window-manager kernel. Syscall vector \$2003-\$204E. Loads here at boot.	~13.5 KB -\$5585
\$5700	OS scratch LINEBUF, CWDPATH (\$5800), the FS / shell / make variables — the half of the band that stayed put.	~0.5 KB -\$58FF
\$5900	TPA — transient program area \$5900 = TPABASE Every RUNnable program loads and executes here. The self-compiled C compiler is a 30.8 KB image that fits with room to spare. A program's frame stack grows down from \$F7FF.	39.8 KB -\$F7FF

\$F800	HISTRING Shell history ring (8×64 B). CSTACKTOP \$F800 is the boundary above it: a launched C program's frame stack starts at \$F7FF and grows down into the TPA.	512 B -\$F9FF
\$FA00	FSDIRBUF Directory-scan / glob sector page (dir, find, glob).	512 B -\$FBFF
\$FC00	RDBUF Shared file-read buffer — every C command's FOPEN buffer (cat, grep, cp, diff, cc, ...), one open file at a time. Not compiler-specific.	512 B -\$FDFF
\$FE00	Stack OS/shell return stack, top \$FEFF (distinct from the program C stack at \$F7FF↓).	256 B -\$FEFF
\$FF00	Memory-mapped I/O I/O ACIA \$FF04 · CF card \$FF10 · multiply-divide \$FF30 · GL graphics \$FF50.	256 B -\$FFFF

REGION TYPE

ROM — firmware

RAM — OS & scratch

Buffers & stack

Reclaimed — the island

TPA — programs

I/O — devices

I/O PAGE \$FF00

ACIA

\$FF04

CF card

\$FF10

MDU

\$FF30

GL gfx

\$FF50

WHAT CHANGED

● ROM 8 KB → 6 KB. The monitor only used ~5.2 KB; the top 2 KB became RAM.

● TPABASE \$6100 → \$5900. +2 KB per program (now ~39.8 KB).

● Scratch → the island. SBUF and the BIOS/FS scratch moved into \$1800–\$1FFF.

● Compiler self-hosts. The extra room lets cc.c compile itself on the machine.

Single source: generators/gen_memmap.py RAM: 2× 62256 Diagram is schematic; exact ranges & sizes are labelled.